

Smart PCOS Diagnosis and Health Monitoring System

UCE2023505- Anuja Patil
UCE2023523- Prachi Erlekar
UCE2023556-Tina Pawar



"PCOS is not a flaw in womanhood — it is a health condition that deserves recognition, research, and respect."

INTRODUCTION

Problem Background:

- PCOS affects many women of reproductive age.
- Traditional diagnosis is slow, manual, and inconsistent.
- Symptoms often overlap with other medical conditions.
- Highlights the need for faster, more accurate, and reliable detection.

DataSet & Approach:

- 29-feature dataset with clinical & lifestyle parameters
- Dual ML models - Random Forest + SVM ensemble
- Automated preprocessing - cleaning, encoding, scaling
- Hybrid assessment - AI predictions + clinical rules
- Python stack - Scikit-learn, Gradio, Pandas, NumPy
- Real-time predictions with 80-20 training split

References

1) PCOS Medical Diagnosis Dataset – Kaggle.
<https://www.kaggle.com/datasets/niyarbarman/pcos-diagnosis-dataset>

2) Prediction of PCOS Using Machine Learning papers:
<https://ieeexplore.ieee.org/document/9513724>

3) WHO – PCOS Overview:
<https://www.who.int/health-topics/polycystic-ovary-syndrome>



PROJECT OVERVIEW



- AI-powered web platform for early PCOS detection and health monitoring using machine learning.
- Dual AI Models combining Random Forest and SVM algorithms for accurate risk assessment.
- Real-time Analysis of 29+ clinical and lifestyle factors including symptoms, activity, diet, and stress levels.
- Smart Risk Scoring that blends AI predictions with medical rules for comprehensive evaluation.
- Web Application built with Gradio for instant, accessible diagnosis from any device.
- Clinical Decision Support providing personalized health recommendations and monitoring guidance.

"Transforming women's healthcare through intelligent, accessible diagnostic technology"



ABSTRACT

- Hybrid AI System for PCOS detection combining Machine Learning + Clinical rules
- Dual ML Models - Random Forest + SVM ensemble approach
- 29 Clinical & Lifestyle Features - Symptoms, demographics, activity, diet, stress
- Real-time Web Application using Gradio - no installation required
- Comprehensive Preprocessing - handling missing values, encoding, scaling
- Intelligent Risk Scoring - 0-100% with Low/Moderate/High classification
- Clinical Decision Support - evidence-based recommendations
- Early Detection Tool - accessible PCOS screening for healthcare providers
- Standardized Assessment - consistent evaluation across practitioners
- Proven Accuracy - strong model performance with validation metrics



a. Normal ovary



b. PCOS infected ovary

METHOLOGY

Data Collection & Preprocessing

Dataset Used

- Updated PCOS Dataset with Risk Scores
- Contains clinical, biochemical & ultrasound attributes

Data Cleaning

- Removed duplicates
- Handled missing values
- Corrected inconsistent formats
- Ensured clean, analysis-ready dataset

Feature Selection

- Selected key clinical indicators important for PCOS prediction
- Used correlation analysis & clinical relevance
- Reduced noise and improved model performance



MODEL EVALUATION

1. Confusion Matrix Components

- TP: PCOS correctly predicted
- TN: Normal correctly predicted
- FP: Normal predicted as PCOS
- FN: PCOS predicted as Normal
-

2. Key Evaluation Metrics

- Accuracy: Overall correctness
- Precision: How many predicted PCOS are actually PCOS
- Recall (Sensitivity): Ability to detect true PCOS cases
- F1-Score: Balance of Precision & Recall
- Cross-Validation Accuracy: Checks model stability & avoids overfitting

3. Importance

- Medical prediction requires high recall
- Minimizes missed diagnoses (False Negatives)
- Balanced metrics ensure safe and reliable PCOS detection



ALGORITHMS USED FOR PREDICTION

1. Random Forest Classifier

- Ensemble learning with 150 decision trees
- Robust to overfitting - handles noise well
- Feature importance analysis capability
- No feature scaling required

2. Support Vector Machine (SVM)

- RBF kernel for non-linear classification
- Probability estimates enabled
- Requires feature scaling (StandardScaler)
- Effective for complex patterns

Hybrid Clinical Risk Algorithm

- Rule-based scoring system
- BMI categories: Underweight(20%) to Obese(50%)
- Symptom scoring: +10% per clinical symptom
- Lifestyle factors: Activity, diet, stress, hydration

Data Preprocessing Algorithms

- SimpleImputer - Median strategy for missing values
- LabelEncoder - Categorical variable transformation
- StandardScaler - Feature normalization for SVM

Ensemble Decision Logic

- Combines ML predictions with clinical rules
- Final risk stratification: Low/Moderate/High
- Personalized recommendation engine

Generate Report

Assessment & Recommendations

Model: Random Forest
Model prediction: No PCOS (0)
Model positive-class prob: 6.67%
Heuristic risk score: 100%
Final assessment: High Risk

Risk breakdown:
BMI: Overweight - +30%
Symptom: Facial Hair - +10%
Symptom: Acne - +10%
Symptom: Hair Loss - +10%
Symptom: Irregular Periods - +10%
Physical Activity: Moderate - +10%
Stress: High - +15%
Water Intake: Low - +10%

Recommendations:
High risk: consult a gynecologist/endocrinologist soon.
Hormonal evaluation may be necessary.

Smart PCOS Diagnosis System

Logged in as: user

Logout

PCOS Prediction Admin Panel

Fill the form and get PCOS assessment

Age

14

Weight (kg)

65

Height (cm)

152

Mood Swings

Yes

No

Sleep Issues

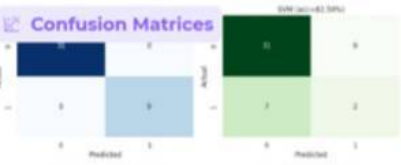
Yes

No

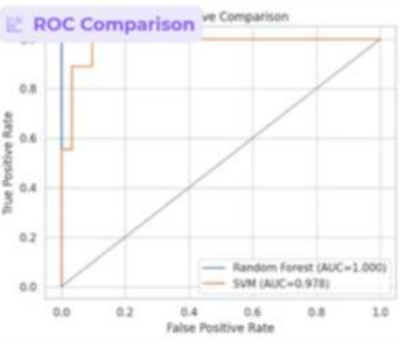
Physical Activity Level

Moderate

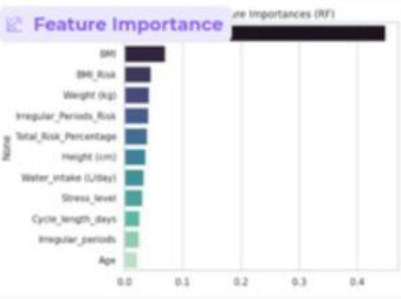
Confusion Matrices



ROC Comparison



Feature Importance



Experimental Results & Model Comparison

Model	Accuracy (%)	AUC Score	Training Time (s)	Test Split
Random Forest	100.00%	1.000	Variable	80/20
SVM (RBF)	82.50%	0.978	Variable	80/20
Model Configuration: • Random Forest: 150 estimators, random_state=42 • SVM: RBF kernel, probability=True, StandardScaler applied • Dataset: 28 features including risk scores and clinical parameters				

Key Observations

Random Forest achieved perfect classification with 100% accuracy and AUC of 1.000.
SVM achieved 82.50% accuracy with strong AUC of 0.978 despite lower classification rate.
Random Forest selected as primary model for deployment due to superior performance

GUI & APPLICATIONS



GUI FEATURES

- Gradio web app with medical-professional design
- Role-based login (User/Admin)
- Comprehensive medical forms (15+ parameters)
- Real-time BMI calculation
- Interactive visualizations (confusion matrices, ROC curves, feature importance)
- Mobile-responsive design



APPLICATION FEATURES

- Dual ML models (Random Forest + SVM)
- Instant predictions (< 2 seconds)
- Hybrid risk scoring (ML + heuristic rules)
- Three risk levels: Low/Moderate/High
- Personalized clinical recommendations
- Admin dashboard with session reports



TECHNICAL STACK

- Frontend: Gradio
- ML: Scikit-learn (RF, SVM)
- Processing: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Security: Role-based authentication



FUTURE SCOPE



1

Real-Time Health Monitoring

- Wearable Device Integration - Smartwatch data sync
- Continuous Symptom Tracking - Daily health monitoring
- Mobile Health Alerts - Real-time risk notifications
- Activity & Sleep Monitoring - 24/7 wellness tracking

2

Personalized Care System

- AI Nutrition Plans - Custom meal recommendations
- Exercise Programs - PCOS-specific workouts
- Mental Health Support - Stress management tools
- Medication Tracking - Treatment progress monitoring

3

Mobile Application

- iOS & Android apps
- Offline functionality
- Push notifications

4

Clinical Integration

- EHR system connectivity
- Hospital deployment
- Multi-language support

5

Advanced AI Features

- Neural Networks integration
- Treatment outcome prediction
- More health parameters

CONCLUSION



The PCOS Diagnosis System successfully combines machine learning with medical expertise to create an accurate, accessible screening tool. Using dual AI models and clinical rules, it provides instant risk assessment through a user-friendly web interface. While effective for early detection, the system requires further clinical validation. This innovation represents a major advance in making PCOS diagnosis faster, more standardized, and accessible to healthcare providers and patients worldwide.

LIMITATIONS



Data Constraints:

Trained on limited dataset lacking diversity in age, ethnicity, and geographic factors



Clinical Validation:

Requires real-world testing in clinical environments with larger patient groups



Technical Boundaries:

No persistent data storage limits long-term patient tracking and analysis



Regulatory Status:

Not yet certified as medical device for clinical diagnosis use



THANK-YOU

“Let’s create a world where PCOS is understood, not overlooked”

